

Structural Resource Theory

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Abstract

We introduce *Structural Resource Theory*, a mathematical framework in which systems are described by hidden allocations, observable states, resources, transformations, and availability boundaries. The primitive object is a structural resource machine

$$\mathcal{S} = (\Theta, X, R, Y, \Sigma, \alpha, \delta, \omega, \chi),$$

where Θ is a space of hidden allocations, X is a state space, R is a resource space, Y is an observation space, $\alpha : \Theta \rightarrow X \times R$ initializes state-resource pairs, $\delta : X \times R \times \Sigma \rightarrow X \times R$ evolves the system, $\omega : X \times R \rightarrow Y$ observes it, and $\chi : R \rightarrow \{0, 1\}$ declares whether resources remain available. Ambiguity Theory appears as the fibre theory of the observation-allocation map $\omega \circ \alpha : \Theta \rightarrow Y$: ambiguity is non-unique resource allocation compatible with the same observation. Exhaustion Theory appears as the zero-boundary dynamics induced by χ : exhaustion is the transition from resource-available states to resource-unavailable states. We prove embedding theorems for ambiguity and exhaustion, define structural resource morphisms, establish categorical closure, develop semiring laws for lifetime composition, provide algorithms for fibre construction, exhaustion detection, ambiguous lifetime computation, and resource-aware learning, and discuss applications in Bayesian inference, finite-state machines, artificial intelligence, machine learning, thermodynamics, computation, economics, finance, and political economy. The resulting theory unifies multiplicity of compatible structures and finitude of admissible continuation.

The treatise ends with “The End”

Contents

1	Introduction	3
2	Preliminaries	3
2.1	Finite State Systems	3
2.2	Resources as Ordered Monoids	3
2.3	Fibres	4
2.4	Well-Founded Orders	4
3	Structural Resource Machines	4
4	Axioms of Structural Resource Theory	5
5	Ambiguity as Fibre Theory	6
6	Bayesian Probability as a Resource-Observation System	6
7	Exhaustion as Resource Boundary Theory	7
8	The Unification Theorem	8

9	Regimes of Structural Resource Theory	9
10	Ambiguous Exhaustion	9
11	Algebra of Structural Resources	10
11.1	Sequential Composition	10
11.2	Choice and Parallel Composition	10
11.3	Min-Plus and Max-Plus Semantics	11
12	Morphisms and the Category of Structural Resource Machines	12
13	Algorithms	13
14	Vector Graphics	15
14.1	The Fundamental Structural Resource Diagram	15
14.2	Ambiguity as Horizontal Fibre	15
14.3	Exhaustion as Vertical Resource Descent	15
14.4	The Parent-Theory Diagram	16
15	Applications	16
15.1	Artificial Intelligence and Machine Learning	16
15.2	Resource-Bounded Computation	17
15.3	Thermodynamics	17
15.4	Economics and Finance	17
15.5	Political Economy and Warfare	17
16	Structural Comparison Tables	18
17	Further Theorems	18
17.1	Identifiability and Ambiguity	18
17.2	Resource Monotonicity	19
17.3	Observation Does Not Create Resources	19
18	Open Problems	20
19	Conclusion	20

List of Figures

1	The fundamental diagram of Structural Resource Theory. Ambiguity lives in inverse fibres of $\omega \circ \alpha$. Exhaustion lives in the resource boundary induced by χ . .	15
2	Ambiguity is non-injectivity of the observation-allocation map. Several hidden allocations are compatible with one observation.	15
3	Exhaustion is descent through available resource levels into the unavailable resource boundary.	15
4	Structural Resource Theory as the parent framework. Ambiguity and exhaustion are two projections of the same resource-governed system.	16

List of Tables

1	The basic regimes of Structural Resource Theory.	9
2	AI and ML interpretations of Structural Resource Theory.	16
3	Structural Resource Theory compared with neighbouring frameworks.	18

4	Basic dictionary of Structural Resource Theory.	18
5	Algebraic laws for structural resource lifetimes.	18

1 Introduction

A large class of mathematical, physical, computational, economic, and epistemic systems can be understood through the following primitive pattern:

allocate resources, observe partially, evolve, consume, and possibly run out.

The same pattern explains two phenomena that at first seem distinct.

First, a system may be *ambiguous*: several hidden allocations or structures may produce the same observation. In this case, the observer sees an output y , but the inverse fibre of hidden causes is non-singleton:

$$(\omega \circ \alpha)^{-1}(y) = \{\theta \in \Theta : \omega(\alpha(\theta)) = y\}.$$

This is the structural origin of ambiguity.

Second, a system may be *exhausting*: it may have admissible future evolution only while some resource remains available. When the resource reaches the unavailable region, continuation terminates:

$$\chi(r) = 0.$$

This is the structural origin of exhaustion.

The purpose of this treatise is to define a parent theory in which both are special cases.

The central slogan is:

Ambiguity is horizontal multiplicity in allocation.

Exhaustion is vertical descent in resource availability.

Ambiguity Theory studies unresolved alternatives. Exhaustion Theory studies finite futures. Structural Resource Theory studies both as aspects of one object: an observed resource-governed system with hidden allocation and finite admissible continuation.

2 Preliminaries

2.1 Finite State Systems

A finite state machine consists of a finite state space Q , an action alphabet Σ , and a transition map

$$\delta : Q \times \Sigma \rightarrow Q.$$

In resource systems, the transition map is generally partial:

$$\delta : Q \times \Sigma \rightharpoonup Q.$$

Partiality means that some attempted continuations are inadmissible.

2.2 Resources as Ordered Monoids

A resource space is often an ordered commutative monoid

$$(R, +, 0, \leq),$$

where 0 is the empty or exhausted resource, $+$ combines resources, and $\leq$ orders resource availability.

Examples include:

$$R = \mathbb{N}, \quad R = \mathbb{N}^k, \quad R = \mathbb{R}_{\geq 0}, \quad R = \Delta^{n-1},$$

where Δ^{n-1} is the probability simplex.

2.3 Fibres

For a map

$$f : A \rightarrow B,$$

the fibre of f over $b \in B$ is

$$f^{-1}(b) = \{a \in A : f(a) = b\}.$$

A non-singleton fibre is the basic mathematical form of underdetermination.

2.4 Well-Founded Orders

A strict order $<$ on a set W is well-founded if there is no infinite descending chain

$$w_0 > w_1 > w_2 > \dots.$$

Well-founded orders are central to termination and exhaustion.

3 Structural Resource Machines

Definition 3.1 (Structural resource machine). A structural resource machine is a tuple

$$\mathcal{S} = (\Theta, X, R, Y, \Sigma, \alpha, \delta, \omega, \chi),$$

where:

1. Θ is a hidden allocation space;
2. X is a state space;
3. R is a resource space;
4. Y is an observation space;
5. Σ is an action or input alphabet;
6. $\alpha : \Theta \rightarrow X \times R$ is an allocation map;
7. $\delta : X \times R \times \Sigma \rightarrow X \times R$ is a partial transition map;
8. $\omega : X \times R \rightarrow Y$ is an observation map;
9. $\chi : R \rightarrow \{0, 1\}$ is an availability predicate.

The allocation map initializes hidden structure:

$$\theta \mapsto (x, r).$$

The observation map reveals only:

$$y = \omega(x, r).$$

The transition map evolves the state-resource pair:

$$(x_t, r_t) \xrightarrow{at} (x_{t+1}, r_{t+1}).$$

The predicate χ determines whether the resource permits continuation.

Definition 3.2 (Available and unavailable resources). Define

$$R^{\text{av}} = \chi^{-1}(1), \quad R^{\text{un}} = \chi^{-1}(0).$$

A resource $r \in R$ is available if $r \in R^{\text{av}}$, and unavailable if $r \in R^{\text{un}}$.

Definition 3.3 (Structural state space). The structural state space of $\mathcal{S}$ is

$$Q = X \times R.$$

The evolving and exhausted regions are

$$Q^{\text{ev}} = X \times R^{\text{av}}, \quad Q^{\text{ex}} = X \times R^{\text{un}}.$$

Therefore:

$$Q = Q^{\text{ev}} \sqcup Q^{\text{ex}}$$

whenever

$$R = R^{\text{av}} \sqcup R^{\text{un}}.$$

4 Axioms of Structural Resource Theory

Axiom 4.1 (Allocation). Every system begins with a hidden allocation:

$$\alpha : \Theta \rightarrow X \times R.$$

Axiom 4.2 (Observation). The observer accesses only

$$y = \omega(x, r),$$

not necessarily x , r , or θ .

Axiom 4.3 (Resource availability). There is a predicate

$$\chi : R \rightarrow \{0, 1\}$$

partitioning resources into available and unavailable regions.

Axiom 4.4 (Resource-governed evolution). Admissible dynamics are defined by a partial transition map

$$\delta : X \times R \times \Sigma \rightharpoonup X \times R.$$

Axiom 4.5 (Consumption). There is an order or preorder on R such that productive transitions weakly consume resources:

$$\delta(x, r, a) = (x', r') \implies r' \leq r.$$

Axiom 4.6 (Ambiguity). Ambiguity occurs when the observation-allocation fibre is non-singleton:

$$|(\omega \circ \alpha)^{-1}(y)| > 1.$$

Axiom 4.7 (Exhaustion). Exhaustion occurs exactly when the resource is unavailable:

$$(x, r) \in Q^{\text{ex}} \iff \chi(r) = 0.$$

Axiom 4.8 (Finite descent). If there exists a well-founded rank

$$\rho : R^{\text{av}} \rightarrow W$$

such that every productive transition strictly decreases ρ , then every productive trajectory exhausts in finite time.

5 Ambiguity as Fibre Theory

Definition 5.1 (Observation-allocation map). The observation-allocation map of a structural resource machine is

$$\Omega = \omega \circ \alpha : \Theta \rightarrow Y.$$

Definition 5.2 (Allocation fibre). For an observation $y \in Y$, the allocation fibre is

$$F_y = \Omega^{-1}(y) = \{\theta \in \Theta : \omega(\alpha(\theta)) = y\}.$$

Definition 5.3 (Structural ambi-object). If

$$F_y = \{\theta_1, \dots, \theta_n\},$$

then the structural ambi-object over y is

$$A_y = \langle \theta_1 \mid \theta_2 \mid \dots \mid \theta_n \rangle.$$

Theorem 5.4 (Ambiguity embedding theorem). *Every structural resource machine induces a family of ambi-objects indexed by observations:*

$$y \mapsto A_y.$$

Moreover, ambiguity occurs exactly at those $y \in Y$ for which

$$|F_y| > 1.$$

Proof. For each observation y , the fibre $F_y = \Omega^{-1}(y)$ is the set of hidden allocations compatible with y . If $F_y = \{\theta_1, \dots, \theta_n\}$, define

$$A_y = \langle \theta_1 \mid \dots \mid \theta_n \rangle.$$

This assignment is well-defined for finite fibres. If $|F_y| = 1$, there is exactly one compatible hidden allocation and no structural ambiguity. If $|F_y| > 1$, more than one hidden allocation produces the same observation, and the observer cannot internally select one of them from y alone. Therefore ambiguity occurs exactly when the fibre is non-singleton. $\square$

Remark 5.5. This theorem does not require probabilities. Probability enters when Θ , R , or Y carries probabilistic structure.

6 Bayesian Probability as a Resource-Observation System

Let

$$\Delta^{n-1} = \left\{ p \in \mathbb{R}_{\geq 0}^n : \sum_{i=1}^n p_i = 1 \right\}$$

be the probability simplex.

Given positive likelihoods

$$\ell_i > 0,$$

the Bayesian update map is

$$B_e : \Delta^{n-1} \rightarrow \Delta^{n-1}, \quad (B_e(\pi))_i = \frac{\ell_i \pi_i}{\sum_{j=1}^n \ell_j \pi_j}.$$

Definition 6.1 (Bayesian structural resource machine). A Bayesian structural resource machine is obtained by taking

$$\Theta = \Delta^{n-1}, \quad X = \{e\}, \quad R = \Delta^{n-1}, \quad Y = Y_B,$$

with

$$\alpha(\pi) = (e, \pi),$$

and an observation map

$$\omega(e, \pi) = \Gamma(B_e(\pi)),$$

where

$$\Gamma : \Delta^{n-1} \rightarrow Y_B$$

is an observational coarse-graining or reporting map.

Remark 6.2. If Γ is the identity and all likelihoods are strictly positive, the ordinary Bayesian update map is generally invertible on the relative interior of the simplex. Nontrivial ambiguity therefore requires coarse-graining, quotienting, loss of information, non-identifiability, or another observation map after Bayesian updating.

Theorem 6.3 (Bayesian ambiguity theorem). *In a Bayesian structural resource machine, the ambiguity fibre over an observed report $y \in Y_B$ is*

$$F_y = \{\pi \in \Delta^{n-1} : \Gamma(B_e(\pi)) = y\}.$$

Thus Bayesian ambiguity is non-identifiability of the prior allocation under the observed posterior report.

Proof. By definition,

$$\Omega(\pi) = \omega(\alpha(\pi)).$$

Since

$$\alpha(\pi) = (e, \pi),$$

we have

$$\Omega(\pi) = \omega(e, \pi) = \Gamma(B_e(\pi)).$$

Therefore

$$F_y = \Omega^{-1}(y) = \{\pi \in \Delta^{n-1} : \Gamma(B_e(\pi)) = y\}.$$

The fibre is non-singleton exactly when more than one prior π gives the same observed posterior report y . This is precisely non-identifiability of the prior allocation. $\square$

7 Exhaustion as Resource Boundary Theory

Definition 7.1 (Induced exhaustion partition). Given a structural resource machine, define

$$Q = X \times R,$$

$$Q^{\text{ev}} = X \times \chi^{-1}(1), \quad Q^{\text{ex}} = X \times \chi^{-1}(0).$$

Theorem 7.2 (Exhaustion embedding theorem). *Every structural resource machine induces an exhaustion machine on the state space*

$$Q = X \times R$$

with evolving region Q^{ev} and exhausted region Q^{ex} .

Proof. Since

$$\chi : R \rightarrow \{0, 1\},$$

we have a partition

$$R = \chi^{-1}(1) \sqcup \chi^{-1}(0).$$

Taking products with X , we obtain

$$X \times R = X \times \chi^{-1}(1) \sqcup X \times \chi^{-1}(0).$$

Define these two components as Q^{ev} and Q^{ex} , respectively. The transition map of the induced exhaustion machine is the partial map

$$\delta : X \times R \times \Sigma \rightharpoonup X \times R$$

restricted to Q^{ev} . Therefore the structural resource machine determines an exhaustion machine. $\square$

Definition 7.3 (Strict resource descent). A structural resource machine has strict resource descent if there exists a well-founded ordered set W and a rank map

$$\rho : R^{\text{av}} \rightarrow W$$

such that whenever

$$\delta(x, r, a) = (x', r')$$

with $r, r' \in R^{\text{av}}$, we have

$$\rho(r') < \rho(r).$$

Theorem 7.4 (Finite exhaustion theorem). *If a structural resource machine has strict resource descent into a well-founded resource rank, then every admissible trajectory remaining in Q^{ev} is finite. Hence every maximal trajectory either reaches Q^{ex} or becomes undefined after finitely many productive transitions.*

Proof. Assume for contradiction that there exists an infinite admissible trajectory

$$(x_0, r_0) \rightarrow (x_1, r_1) \rightarrow (x_2, r_2) \rightarrow \dots$$

remaining in Q^{ev} . Since each transition is productive and resource-available, strict resource descent gives

$$\rho(r_0) > \rho(r_1) > \rho(r_2) > \dots$$

This is an infinite descending chain in W , contradicting well-foundedness. Therefore no infinite productive trajectory remains in Q^{ev} . Every maximal trajectory therefore reaches Q^{ex} or becomes undefined in finite time. $\square$

8 The Unification Theorem

Theorem 8.1 (Structural resource unification theorem). *Let*

$$\mathcal{S} = (\Theta, X, R, Y, \Sigma, \alpha, \delta, \omega, \chi)$$

be a structural resource machine. Then:

1. *The fibres of $\omega \circ \alpha : \Theta \rightarrow Y$ define ambi-objects.*
2. *The predicate $\chi : R \rightarrow \{0, 1\}$ defines an exhaustion partition of $X \times R$.*

3. If resource descent is strict in a well-founded order, then exhaustion occurs in finite time.

Therefore Ambiguity Theory and Exhaustion Theory are respectively the fibre theory and boundary-dynamics theory of Structural Resource Theory.

Proof. Part 1 is the ambiguity embedding theorem. Part 2 is the exhaustion embedding theorem. Part 3 is the finite exhaustion theorem. Since all three structures are induced from the single tuple $\mathcal{S}$, ambiguity and exhaustion arise as two projections of the same structural resource system: one through inverse images of observations, the other through the resource availability boundary. $\square$

9 Regimes of Structural Resource Theory

For an observation y , define

$$F_y = (\omega \circ \alpha)^{-1}(y).$$

For a resource r , define availability by $\chi(r)$.

Regime	Fibre condition	Resource condition	Interpretation
Determinate active	$ F_y = 1$	$\chi(r) = 1$	Unique cause, continuation possible
Ambiguous active	$ F_y > 1$	$\chi(r) = 1$	Multiple causes, continuation possible
Determinate exhausted	$ F_y = 1$	$\chi(r) = 0$	Unique cause, continuation impossible
Ambiguous exhausted	$ F_y > 1$	$\chi(r) = 0$	Multiple causes, continuation impossible
Impossible observation	$ F_y = 0$	any	Inconsistency or model failure

Table 1: The basic regimes of Structural Resource Theory.

10 Ambiguous Exhaustion

Definition 10.1 (Ambiguous resource state). An ambiguous resource state is an ambi-object over $X \times R$:

$$A = \langle (x_1, r_1) \mid \cdots \mid (x_n, r_n) \rangle.$$

Definition 10.2 (Branch lifetime). Let $L(x, r)$ be the remaining productive lifetime of state-resource pair (x, r) . The ambiguous lifetime profile of

$$A = \langle (x_1, r_1) \mid \cdots \mid (x_n, r_n) \rangle$$

is

$$\Lambda(A) = \langle L(x_1, r_1) \mid \cdots \mid L(x_n, r_n) \rangle.$$

Definition 10.3 (Existential and universal ambiguous exhaustion time). For an ambiguous resource state A , define

$$\tau_{\exists}(A) = \min_i L(x_i, r_i),$$

and

$$\tau_{\forall}(A) = \max_i L(x_i, r_i).$$

Theorem 10.4 (Ambiguous lifetime theorem). *Let*

$$A = \langle (x_1, r_1) \mid \cdots \mid (x_n, r_n) \rangle$$

be an ambiguous resource state whose branches each have finite lifetime. Then:

$$\tau_{\exists}(A) = \min_i L(x_i, r_i)$$

is the first time at which at least one branch exhausts, and

$$\tau_{\forall}(A) = \max_i L(x_i, r_i)$$

is the first time by which all branches have exhausted.

Proof. Branch i exhausts after exactly $L(x_i, r_i)$ productive transitions. The earliest branch exhaustion therefore occurs at the minimum of the branch lifetimes:

$$\min_i L(x_i, r_i).$$

All branches have exhausted only once the longest-lived branch has exhausted. Therefore the all-branches exhaustion time is

$$\max_i L(x_i, r_i).$$

□

11 Algebra of Structural Resources

11.1 Sequential Composition

Let two deterministic structural resource processes have lifetimes τ_1 and τ_2 . Sequential composition means running the second after the first.

Proposition 11.1 (Sequential lifetime law). *For deterministic finite-horizon processes,*

$$\tau(\mathcal{S}_1; \mathcal{S}_2) = \tau(\mathcal{S}_1) + \tau(\mathcal{S}_2).$$

Proof. The composed process first spends $\tau(\mathcal{S}_1)$ transitions in $\mathcal{S}_1$, then $\tau(\mathcal{S}_2)$ transitions in $\mathcal{S}_2$. Hence the total number of productive transitions is the sum. □

11.2 Choice and Parallel Composition

Proposition 11.2 (Choice lifetime law). *For deterministic processes,*

$$\tau_{\text{worst}}(\mathcal{S}_1 \oplus \mathcal{S}_2) = \min\{\tau(\mathcal{S}_1), \tau(\mathcal{S}_2)\},$$

and

$$\tau_{\text{best}}(\mathcal{S}_1 \oplus \mathcal{S}_2) = \max\{\tau(\mathcal{S}_1), \tau(\mathcal{S}_2)\}.$$

Proof. Under worst-case choice, the selected branch may be the one with shorter lifetime. Under best-case choice, the selected branch may be the one with longer lifetime. Thus the two horizons are respectively the minimum and maximum of the two lifetimes. □

Proposition 11.3 (Parallel lifetime law). *For deterministic parallel processes:*

$$\tau_{\text{fragile}}(\mathcal{S}_1 \otimes \mathcal{S}_2) = \min\{\tau(\mathcal{S}_1), \tau(\mathcal{S}_2)\},$$

if the system fails when any component exhausts, while

$$\tau_{\text{redundant}}(\mathcal{S}_1 \otimes \mathcal{S}_2) = \max\{\tau(\mathcal{S}_1), \tau(\mathcal{S}_2)\},$$

if the system fails only when all components exhaust.

Proof. In fragile parallelism, the first exhausted component terminates the composite system, giving the minimum lifetime. In redundant parallelism, the composite survives until the last component exhausts, giving the maximum lifetime. □

11.3 Min-Plus and Max-Plus Semantics

Let

$$\mathbb{N}_\infty = \mathbb{N} \cup \{\infty\}.$$

Theorem 11.4 (Pessimistic structural resource semiring). *The structure*

$$(\mathbb{N}_\infty, \min, +)$$

is an idempotent semiring with additive identity ∞ and multiplicative identity 0.

Proof. The operation $\min$ is associative, commutative, and idempotent:

$$\min(a, a) = a.$$

Its identity is ∞ , since

$$\min(a, \infty) = a.$$

Addition is associative and has identity 0. Addition distributes over minimum:

$$a + \min(b, c) = \min(a + b, a + c).$$

Finally, ∞ is absorbing for addition:

$$a + \infty = \infty.$$

Therefore $(\mathbb{N}_\infty, \min, +)$ is an idempotent semiring. □

Definition 11.5 (Max-plus domain). Let

$$\mathbb{N}_{\pm\infty} = \mathbb{N} \cup \{-\infty, \infty\}.$$

The expression $\infty + (-\infty)$ is left undefined.

Theorem 11.6 (Optimistic structural resource semiring). *On the defined extended-natural expressions,*

$$(\mathbb{N}_{\pm\infty}, \max, +)$$

is an idempotent semiring with additive identity $-\infty$ and multiplicative identity 0.

Proof. The operation $\max$ is associative, commutative, and idempotent:

$$\max(a, a) = a.$$

Its identity is $-\infty$, since

$$\max(a, -\infty) = a.$$

Addition is associative where defined and has identity 0. Addition distributes over maximum:

$$a + \max(b, c) = \max(a + b, a + c).$$

The additive zero $-\infty$ is absorbing for addition:

$$a + (-\infty) = -\infty.$$

Thus the max-plus structure is an idempotent semiring on its defined domain. □

12 Morphisms and the Category of Structural Resource Machines

Definition 12.1 (Structural resource morphism). Let

$$\mathcal{S} = (\Theta, X, R, Y, \Sigma, \alpha, \delta, \omega, \chi)$$

and

$$\mathcal{S}' = (\Theta', X', R', Y', \Sigma, \alpha', \delta', \omega', \chi')$$

be structural resource machines over the same action alphabet. A structural resource morphism

$$f : \mathcal{S} \rightarrow \mathcal{S}'$$

is a quadruple

$$f = (f_\Theta, f_X, f_R, f_Y)$$

where

$$f_\Theta : \Theta \rightarrow \Theta', \quad f_X : X \rightarrow X', \quad f_R : R \rightarrow R', \quad f_Y : Y \rightarrow Y',$$

such that:

$$\begin{aligned} (f_X \times f_R) \circ \alpha &= \alpha' \circ f_\Theta, \\ f_Y \circ \omega &= \omega' \circ (f_X \times f_R), \\ \chi' \circ f_R &= \chi, \end{aligned}$$

and whenever

$$\delta(x, r, a) = (x', r'),$$

we have

$$\delta'(f_X(x), f_R(r), a) = (f_X(x'), f_R(r')).$$

Theorem 12.2 (Category of structural resource machines). *Structural resource machines over a fixed action alphabet, together with structural resource morphisms, form a category $\mathbf{SRes}_\Sigma$.*

Proof. For every structural resource machine $\mathcal{S}$, the identity quadruple

$$(\text{id}_\Theta, \text{id}_X, \text{id}_R, \text{id}_Y)$$

satisfies all four morphism conditions, so identities exist.

Now let

$$f : \mathcal{S} \rightarrow \mathcal{S}', \quad g : \mathcal{S}' \rightarrow \mathcal{S}''$$

be structural resource morphisms. Their composite is

$$g \circ f = (g_\Theta \circ f_\Theta, g_X \circ f_X, g_R \circ f_R, g_Y \circ f_Y).$$

Allocation preservation follows from

$$(g_X \times g_R) \circ (f_X \times f_R) \circ \alpha = (g_X \times g_R) \circ \alpha' \circ f_\Theta = \alpha'' \circ g_\Theta \circ f_\Theta.$$

Observation preservation follows from

$$g_Y \circ f_Y \circ \omega = g_Y \circ \omega' \circ (f_X \times f_R) = \omega'' \circ (g_X \times g_R) \circ (f_X \times f_R).$$

Availability preservation follows from

$$\chi'' \circ g_R \circ f_R = \chi' \circ f_R = \chi.$$

Transition preservation follows by applying first f and then g to any admissible transition. Function composition is associative. Hence $\mathbf{SRes}_\Sigma$ is a category. $\square$

Proposition 12.3 (Fibre functor). *For every structural resource machine $\mathcal{S}$, the assignment*

$$y \mapsto F_y = (\omega \circ \alpha)^{-1}(y)$$

defines the ambiguity component of $\mathcal{S}$.

Proof. This is exactly the fibre construction of the observation-allocation map. Each observation y is assigned the set of hidden allocations compatible with it. $\square$

Proposition 12.4 (Boundary functor). *For every structural resource machine $\mathcal{S}$, the assignment*

$$\mathcal{S} \mapsto (X \times R, \Sigma, \delta, Q^{\text{ev}}, Q^{\text{ex}})$$

defines the exhaustion component of $\mathcal{S}$.

Proof. The resource predicate χ induces the partition

$$X \times R = X \times \chi^{-1}(1) \sqcup X \times \chi^{-1}(0).$$

Together with the transition map δ , this is precisely an exhaustion machine. $\square$

13 Algorithms

Algorithms are stated using ordinary mathematical tables rather than specialized algorithm packages.

Algorithm 13.1 (Constructing structural ambiguity fibres). Input: finite Θ , allocation map α , observation map ω , and observation y .

Step	Operation
1	Set $F_y = \emptyset$.
2	For each $\theta \in \Theta$, compute $(x, r) = \alpha(\theta)$.
3	Compute $y_\theta = \omega(x, r)$.
4	If $y_\theta = y$, insert θ into F_y .
5	Return $A_y = \langle \theta : \theta \in F_y \rangle$.

Correctness. The algorithm tests each $\theta \in \Theta$ and includes it exactly when

$$\omega(\alpha(\theta)) = y.$$

Therefore the returned set is exactly

$$F_y = (\omega \circ \alpha)^{-1}(y).$$

Writing this fibre as an ambi-object gives A_y . $\square$

Algorithm 13.2 (Detecting finite exhaustion). Input: finite X , finite R , transition map δ , availability predicate χ , initial state (x_0, r_0) .

Step	Operation
1	Build the directed graph on $Q = X \times R$.
2	Compute reachable states from (x_0, r_0) .
3	Restrict to $Q^{\text{ev}} = X \times \chi^{-1}(1)$.
4	Check whether the reachable evolving subgraph contains a directed cycle.
5	If no cycle exists, return “finite exhaustion”.
6	If a cycle exists, return “possible non-exhaustion”.

Correctness. In a finite directed graph, an infinite trajectory exists if and only if some reachable directed cycle exists. If the reachable evolving subgraph has no directed cycle, then every evolving trajectory is finite and must reach Q^{ex} or become undefined. If such a cycle exists, following it gives a non-exhausting run under the corresponding action labels. $\square$

Algorithm 13.3 (Ambiguous lifetime profile). Input: ambiguous state

$$A = \langle (x_1, r_1) \mid \cdots \mid (x_n, r_n) \rangle.$$

Step	Operation
1	For each branch i , compute $L_i = L(x_i, r_i)$.
2	Return $\Lambda(A) = \langle L_1 \mid \cdots \mid L_n \rangle$.
3	Compute $\tau_{\exists}(A) = \min_i L_i$.
4	Compute $\tau_{\forall}(A) = \max_i L_i$.

Correctness. By definition, the ambiguous lifetime profile records the remaining lifetime of each branch. The first branch to exhaust has lifetime $\min_i L_i$. The last branch to exhaust has lifetime $\max_i L_i$. Therefore the algorithm returns the correct existential and universal ambiguous exhaustion times. $\square$

Algorithm 13.4 (Resource-aware hypothesis filtering). Input: hypothesis class H , data stream $D_1, D_2, \dots$, resource budget r_0 , loss threshold ϵ .

Step	Operation
1	Initialize $V_0 = H$, $r = r_0$.
2	For $t = 1, 2, \dots$ while $r > 0$:
3	Observe batch D_t .
4	Set $V_t = \{h \in V_{t-1} : \ell(h, D_t) \leq \epsilon\}$.
5	Update resource $r \leftarrow r - c(V_{t-1}, D_t)$.
6	If $ V_t = 0$, return inconsistent exhaustion.
7	If $ V_t = 1$, return resolved hypothesis.
8	If $r = 0$, return exhausted ambiguity $\langle h : h \in V_t \rangle$.

Correctness. At each step the algorithm retains exactly those hypotheses satisfying the loss constraint on the newly observed data. Thus V_t is the version space after t batches. The resource budget is decreased by the declared cost function. If $V_t = \emptyset$, no hypothesis remains compatible with the observations. If $|V_t| = 1$, ambiguity is resolved. If the resource reaches zero while $|V_t| > 1$, the process terminates with a non-singleton compatible hypothesis fibre, hence with exhausted ambiguity. $\square$

14 Vector Graphics

14.1 The Fundamental Structural Resource Diagram

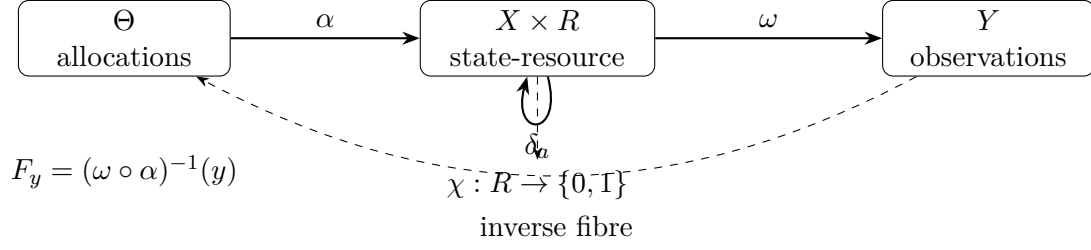


Figure 1: The fundamental diagram of Structural Resource Theory. Ambiguity lives in inverse fibres of $\omega \circ \alpha$. Exhaustion lives in the resource boundary induced by χ .

14.2 Ambiguity as Horizontal Fibre

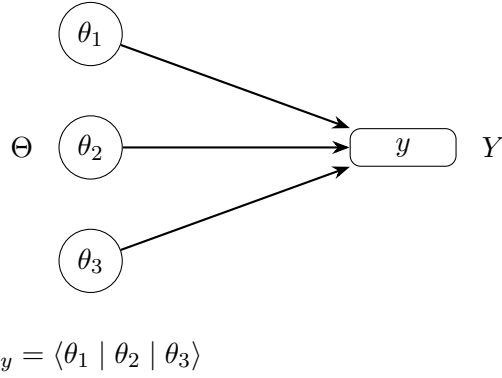


Figure 2: Ambiguity is non-injectivity of the observation-allocation map. Several hidden allocations are compatible with one observation.

14.3 Exhaustion as Vertical Resource Descent

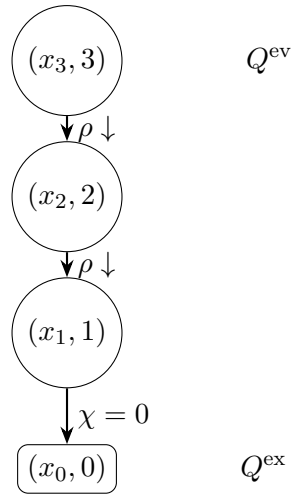


Figure 3: Exhaustion is descent through available resource levels into the unavailable resource boundary.

14.4 The Parent-Theory Diagram

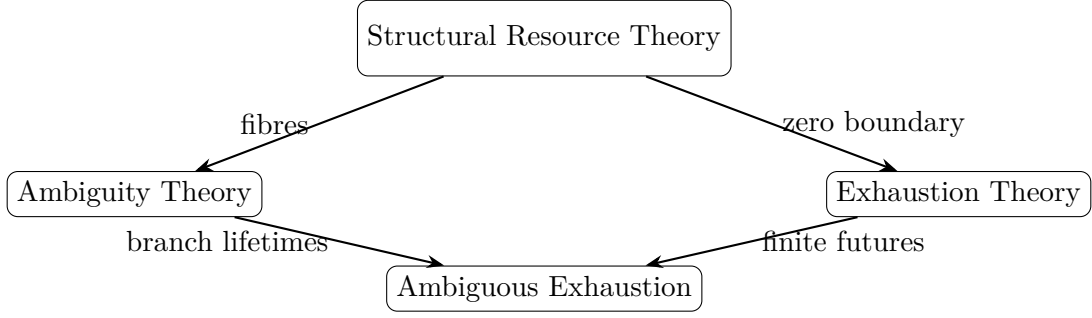


Figure 4: Structural Resource Theory as the parent framework. Ambiguity and exhaustion are two projections of the same resource-governed system.

15 Applications

15.1 Artificial Intelligence and Machine Learning

Structural Resource Theory gives a compact language for several recurring AI and ML patterns.

Let:

$$\Theta = H$$

be a hypothesis class,

$$X$$

be the model state,

$$R$$

be computational, sample, memory, or token budget, and

$$Y$$

be observed training or validation behaviour.

Ambiguity appears when many hypotheses produce the same observation:

$$F_y = \{h \in H : \omega(\alpha(h)) = y\}.$$

Exhaustion appears when budget reaches zero:

$$\chi(r) = 0.$$

AI/ML setting	Hidden allocation Θ	Resource R	Exhaustion event
Version-space learning	Hypotheses	Samples or labels	No budget or no hypotheses
Decision trees	Candidate splits	Depth/sample mass	No valid split
Reinforcement learning	Policies/models	Energy/time/safety budget	No safe admissible action
Large language models	Latent continuation states	Context/tokens	Token or context exhaustion
Neural architecture search	Architectures	Search budget	Search frontier depleted
Bayesian inference	Priors	Probability mass/report precision	Non-identifiability or coarse report

Table 2: AI and ML interpretations of Structural Resource Theory.

15.2 Resource-Bounded Computation

A bounded computation may be represented by:

$$R = \mathbb{N}^k,$$

where the coordinates count time, memory, stack depth, communication, or oracle calls. A transition is admissible only when the required resource exists.

The exhausted state is not necessarily accepting or rejecting. It may mean:

out of memory, out of time, out of stack, out of communication budget.

15.3 Thermodynamics

In thermodynamics:

Θ = microstates, X = macrostates, R = available free energy, Y = observable thermodynamic data.

Ambiguity is the multiplicity of microstates compatible with the same macrostate. Exhaustion is the loss of usable free energy.

Thus entropy-like multiplicity and energy-like depletion are unified:

entropy counts hidden compatible allocations; free energy bounds continuation.

15.4 Economics and Finance

In economics and finance:

Θ = hidden balance-sheet allocations,

X = observable institutional state,

R = liquidity, collateral, credit, labour, energy, trust,

Y = market prices, spreads, ratings, reported statistics.

Ambiguity arises when multiple hidden structures generate the same market signal. Exhaustion arises when liquidity, collateral, trust, or political capacity reaches zero.

A crisis may be modeled as:

crisis = ambiguous allocation plus accelerating resource exhaustion.

15.5 Political Economy and Warfare

In political economy and warfare:

R

may represent manpower, ammunition, legitimacy, logistics, attention, industrial capacity, or alliance credibility. Ambiguity appears when adversaries cannot infer the true allocation of resources. Exhaustion appears when one side's resource frontier is depleted.

This gives a formal distinction between:

uncertainty about the opponent's allocation

and

actual exhaustion of the opponent's continuation capacity.

16 Structural Comparison Tables

Theory	Primitive object	Main structure	Failure/terminal event
Probability theory	Distribution	Weighted uncertainty	Conditioning or updating
Ambiguity Theory	Ambi-object	Non-singleton fibre	Resolution
Exhaustion Theory	Exhaustion machine	Evolving/exhausted partition	Termination
Resource theory	Resource state	Convertibility pre-order	Inaccessibility
Structural Resource Theory	Resource machine	Fibre plus boundary	Ambiguous exhaustion

Table 3: Structural Resource Theory compared with neighbouring frameworks.

Component	Mathematical object	Role
Hidden allocation	Θ	Latent cause or initial resource structure
State	X	Internal system configuration
Resource	R	Continuation capacity
Observation	Y	Externally visible report
Allocation map	$\alpha : \Theta \rightarrow X \times R$	Initialization
Observation map	$\omega : X \times R \rightarrow Y$	Partial visibility
Transition map	δ	Resource-consuming evolution
Availability predicate	$\chi : R \rightarrow \{0, 1\}$	Exhaustion boundary
Ambiguity fibre	F_y	Compatible hidden allocations
Lifetime	$L(x, r)$	Remaining productive continuation

Table 4: Basic dictionary of Structural Resource Theory.

Operation	Lifetime law	Interpretation
Sequential composition	$\tau_1 + \tau_2$	Run one process after another
Worst-case choice	$\min(\tau_1, \tau_2)$	Adversarial branch selection
Best-case choice	$\max(\tau_1, \tau_2)$	Optimistic branch selection
Fragile parallelism	$\min(\tau_1, \tau_2)$	Any component failure ends system
Redundant parallelism	$\max(\tau_1, \tau_2)$	All components must fail
Ambiguous lifetime	$\langle \tau_1 \dots \tau_n \rangle$	Branchwise finite futures

Table 5: Algebraic laws for structural resource lifetimes.

17 Further Theorems

17.1 Identifiability and Ambiguity

Definition 17.1 (Structural identifiability). A structural resource machine is identifiable at observation y if

$$|F_y| = 1.$$

It is globally identifiable if this holds for every $y \in \text{im}(\omega \circ \alpha)$.

Theorem 17.2 (Identifiability theorem). *A structural resource machine is globally identifiable if and only if the observation-allocation map*

$$\Omega = \omega \circ \alpha$$

is injective.

Proof. Suppose Ω is injective. Then for every $y \in \text{im}(\Omega)$, there is exactly one $\theta \in \Theta$ such that $\Omega(\theta) = y$. Hence $|F_y| = 1$.

Conversely, suppose every fibre over the image has size one. If

$$\Omega(\theta_1) = \Omega(\theta_2) = y,$$

then $\theta_1, \theta_2 \in F_y$. Since $|F_y| = 1$, we have $\theta_1 = \theta_2$. Therefore Ω is injective. $\square$

17.2 Resource Monotonicity

Definition 17.3 (Resource monotone). A function

$$M : X \times R \rightarrow \mathbb{R}$$

is a resource monotone if every admissible transition satisfies

$$M(x', r') \leq M(x, r).$$

It is strict if the inequality is strict for productive transitions.

Proposition 17.4 (Monotone obstruction). *If there exists a strict resource monotone M taking values in a finite well-ordered set, then the system has no infinite productive trajectory.*

Proof. An infinite productive trajectory would give an infinite strictly decreasing sequence of values of M . A finite well-ordered set admits no such sequence. Therefore no infinite productive trajectory exists. $\square$

17.3 Observation Does Not Create Resources

Definition 17.5 (Resource-forgetting observation). An observation map $\omega : X \times R \rightarrow Y$ is resource-forgetting if there exist distinct resources $r \neq r'$ and states x, x' such that

$$\omega(x, r) = \omega(x', r').$$

Proposition 17.6 (Resource ambiguity). *If $\omega \circ \alpha$ is non-injective because ω forgets resource coordinates, then there exists an observation y for which compatible allocations have distinct resource levels.*

Proof. Since $\omega \circ \alpha$ is non-injective, there exist $\theta_1 \neq \theta_2$ such that

$$\omega(\alpha(\theta_1)) = \omega(\alpha(\theta_2)) = y.$$

Write

$$\alpha(\theta_1) = (x_1, r_1), \quad \alpha(\theta_2) = (x_2, r_2).$$

If non-injectivity is due to forgetting resource coordinates, then $r_1 \neq r_2$. Thus the same observation y is compatible with distinct resource levels. $\square$

Corollary 17.7 (Ambiguous lifetime from resource ambiguity). *If compatible allocations have distinct resource levels and lifetime depends nontrivially on resource level, then the observation has an ambiguous lifetime profile.*

Proof. Let

$$\alpha(\theta_i) = (x_i, r_i)$$

with $r_1 \neq r_2$. If lifetime depends nontrivially on resource level, then for some such pair,

$$L(x_1, r_1) \neq L(x_2, r_2).$$

Thus

$$\Lambda(A_y) = \langle L(x_1, r_1) \mid L(x_2, r_2) \mid \cdots \rangle$$

contains distinct branches. □

18 Open Problems

- P1. Classification.** Classify structural resource machines up to structural resource isomorphism. Determine which invariants are complete.
- P2. Fibre-boundary duality.** Develop a duality between observation fibres and resource boundaries. Determine when ambiguity predicts exhaustion.
- P3. Topos structure.** Determine whether suitable categories of structural resource machines form topoi or fibred categories with useful internal logics.
- P4. Continuous resources.** Extend the finite theory to continuous resource spaces such as $\mathbb{R}_{\geq 0}^n$, probability simplices, manifolds, and thermodynamic state spaces.
- P5. Stochastic structural resources.** Develop Markov and Bayesian versions of Structural Resource Theory with expected exhaustion times and posterior ambiguity fibres.
- P6. Ambiguous control.** Study optimal control when the current resource state is ambiguous. Characterize policies that maximize $\tau_{\exists}$, $\tau_{\forall}$, or risk-sensitive lifetime functionals.
- P7. Learning theory.** Develop generalization bounds in which sample complexity is interpreted as resource exhaustion of a hypothesis fibre.
- P8. Crisis theory.** Formalize economic, financial, and geopolitical crises as simultaneous growth of ambiguity fibres and collapse of resource ranks.
- P9. Algorithmic complexity.** Classify the complexity of fibre computation, lifetime computation, ambiguous lifetime aggregation, and optimal resource allocation.
- P10. Thermodynamic semantics.** Relate structural resource ranks to entropy, free energy, and irreversible computation.

19 Conclusion

Structural Resource Theory unifies two apparently distinct phenomena.

Ambiguity arises when many hidden allocations produce the same observation:

$$|(\omega \circ \alpha)^{-1}(y)| > 1.$$

Exhaustion arises when resource availability reaches the zero or unavailable boundary:

$$\chi(r) = 0.$$

The primitive object is therefore neither a probability distribution, nor an ambi-object, nor a finite state machine alone. It is the structural resource machine:

$$\mathcal{S} = (\Theta, X, R, Y, \Sigma, \alpha, \delta, \omega, \chi).$$

Ambiguity Theory is recovered by taking inverse fibres of $\omega \circ \alpha$. Exhaustion Theory is recovered by taking the partition of $X \times R$ induced by χ . Their joint theory studies ambiguous finite futures:

$$\Lambda(A_y) = \langle \tau_1 \mid \cdots \mid \tau_n \rangle.$$

Thus the central theorem of the treatise is:

Ambiguity and exhaustion are two projections of structural resource dynamics.

The final slogan is:

Structure allocates; observation forgets; dynamics consumes; exhaustion terminates.

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Glossary

Allocation

The hidden initialization of a system, represented by a map $\alpha : \Theta \rightarrow X \times R$.

Allocation fibre

The set

$$F_y = (\omega \circ \alpha)^{-1}(y)$$

of hidden allocations compatible with an observation y .

Ambiguous exhaustion

A state in which several hidden allocations are compatible with the observation and the resource state is exhausted or has branch-dependent exhaustion time.

Ambiguous lifetime profile

The ambi-object

$$\Lambda(A) = \langle L(x_1, r_1) \mid \cdots \mid L(x_n, r_n) \rangle$$

recording branchwise remaining lives.

Ambiguity

Non-singleton compatibility of hidden allocations with one observation:

$$|F_y| > 1.$$

Availability predicate

A map

$$\chi : R \rightarrow \{0, 1\}$$

declaring whether a resource is available or unavailable.

Boundary dynamics

The part of Structural Resource Theory that studies trajectories reaching the unavailable resource boundary.

Exhaustion

The event that the resource becomes unavailable:

$$\chi(r) = 0.$$

Exhaustion partition

The decomposition

$$X \times R = Q^{\text{ev}} \sqcup Q^{\text{ex}},$$

where $Q^{\text{ev}} = X \times \chi^{-1}(1)$ and $Q^{\text{ex}} = X \times \chi^{-1}(0)$.

Fibre theory

The part of Structural Resource Theory that studies inverse images of observations under $\omega \circ \alpha$.

Hidden allocation space

The set Θ of latent causes, priors, configurations, hypotheses, microstates, or initial resource allocations.

Observation

The externally available report

$$y = \omega(x, r).$$

Observation-allocation map

The composite map

$$\Omega = \omega \circ \alpha : \Theta \rightarrow Y.$$

Resource

A continuation-enabling quantity, stock, capacity, budget, or structure represented by an element $r \in R$.

Resource monotone

A function that does not increase along admissible transitions.

Resource rank

A well-founded measure of resource availability used to prove finite exhaustion.

State-resource pair

An element

$$(x, r) \in X \times R$$

describing both internal state and available resource.

Structural resource machine

The tuple

$$\mathcal{S} = (\Theta, X, R, Y, \Sigma, \alpha, \delta, \omega, \chi)$$

that defines the basic object of Structural Resource Theory.

Structural resource morphism

A quadruple of maps preserving allocation, observation, resource availability, and dynamics.

Universal ambiguous exhaustion time

The time

$$\tau_{\forall}(A) = \max_i L(x_i, r_i)$$

by which every branch of an ambiguous resource state has exhausted.

Existential ambiguous exhaustion time

The time

$$\tau_{\exists}(A) = \min_i L(x_i, r_i)$$

at which at least one branch of an ambiguous resource state has exhausted.

The End